Same-disorder curvature in the Slater selected algebra

Fixed-word source calculus, central pinching and covariant observables

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Abstract

We construct all fixed mixed moments of the actual quadratic curvature field with the Hamiltonian, local size fluctuation and Slater-selected operators. The construction retains the original Gaussian disorder. Separate empty and occupied local matrix elements extend the existing nilpotent loss-source calculus to selected chords. Without selected chords the source exponent agrees exactly with the prior tracial one, which proves the central-sector moment and pinching statements directly. The polynomial orbit-density estimate transfers these errors to arbitrary fixed selected polynomial preparations. It also makes the covariant three-field algebra a bounded normal representation on the enlarged GNS space. A nonzero curvature/double-selected overlap rules out an independent-probe substitution in this state.

1 Microscopic definitions and the result

Keep standard CAR, overlapping Hermitian complex Gaussian p-body couplings, the same J in every field, a fixed half-filled Slater vector O, and $\varphi_N = \mathbb{E}\langle O|\cdot|O\rangle$. Let $m = N/2$, $\lambda_N = p^2/N \to \lambda > 0$, $q_N = e^{-4\lambda_N}$, $C_Q = \mathbb{E}H^2$, and C_m its scalar value on $\bigwedge^m \mathbb{C}^N$. All limits fix the word and λ first. Set

$$DX = QX - \sum_i c_i^\dagger X c_i, \qquad \mathcal{L}X = \frac{NX - \sum_i (c_i^\dagger X c_i + c_i X c_i^\dagger)}{2}, \tag{1}$$

$$Z_2 = H^2 - C_Q, \qquad \kappa_N = 2\lambda_N \frac{1-q_N}{1+q_N},$$

$$j_N = C_m^{-1}(2p - D - \kappa_N) Z_2\big|_m, \qquad v_N = 4\lambda_N + \frac{16\lambda_N^2 q_N}{1+q_N}, \tag{2}$$

$$h_N = H_m/\sqrt{C_m}, \qquad W_N = (p - \mathcal{L} - \lambda_N)H\big|_m/\sqrt{C_m},$$

$$u_N = W_N/\sqrt{\lambda_N}, \qquad x_N = j_N/\sqrt{v_N}.$$

The loss maps act on full-Fock operators *before* sector restriction. The deterministic C_m^{-1} is applied afterwards; replacing H by the sector-dependent $H/\sqrt{C_Q}$ before applying D would change the field. $M_N^\dagger$ is the original normalized occupied-to-empty p-particle creation block, with $\varphi_N(M_N M_N^\dagger) = 1$. These are dimensionless string-energy observables, with $\hbar = 1$.

Theorem 1. *Every fixed word in $h_N, W_N, j_N, M_N, M_N^\dagger$ has the limit specified by the finite diagram rule in Sections 2–3. The limiting functional is positive. Its covariant restriction (h, u, x) has the existing tracial mixed-q law with lengths $(1, 1, 2)$. Exact energy pinching of x_N vanishes in every fixed central Schatten norm and on every fixed polynomial preparation in this full family.*

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2 A local source rule with the selected state retained

Expand j into two adjacent Hamiltonian slots with a whole-block D source; expand W into one slot with an $\mathcal{L}$ source. Gaussian Wick centering of Z_2 forbids pairs internal to its two-slot block. Other Hamiltonian slots are ordinary. Write a Gaussian pairing as $\pi = \{(a_e, b_e) : a_e < b_e\}_{e=1}^{k}$. At the right endpoint use $C_{P_e}^{\dagger} C_{Q_e}$, and at the left its adjoint. An unrestricted edge is G. A forward selected edge is F, requiring P_e empty and Q_e occupied in O. A backward selected edge has the reverse constraint. Two creations or two annihilations cannot pair.

Use one nilpotent variable z_a, $z_a^2 = 0$, for each marked block. For a one-site matrix $X = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, the local maps are

$$D_i X = \begin{pmatrix} 0 & b/2 \\ c/2 & d-a \end{pmatrix}, \qquad \mathcal{L}_i X = \frac{1}{2} \begin{pmatrix} a-d & b \\ c & d-a \end{pmatrix}. \tag{3}$$

For membership $S \subset \{P_e, Q_e\}_{e=1}^{k}$, multiply the local CAR matrices inside each block, apply $1 - z_a D_i$ or $1 - z_a \mathcal{L}_i$, and multiply the blocks in their original order. Take the (o, o) entry for $o = 0, 1$, rather than the local trace. Multiply by the Jordan–Wigner regrouping sign

$$(-1)^{\sum_{e<f:\, e \text{ crosses } f} t_e(S) t_f(S)}, \qquad t_e(S) = |S \cap \{P_e, Q_e\}|. \tag{4}$$

Set the answer to zero for any violated selected-label restriction. Call the resulting polynomial $f_o(S; z)$.

Each label has one correct singleton occupancy: P_e at $o = 0$, Q_e at $o = 1$. For a forward diagram its singleton polynomial $s_i(z)$ has constant term one. Because no marked block contains both endpoints of a Gaussian pair, each incident singleton contributes $1 - z_a/2$ to that block. For a block of d_a slots,

$$e_{\text{nil}}^{p \sum_a d_a z_a} \prod_i s_i(z)^p = 1. \tag{5}$$

This removes the extensive terms of $pd_a - D$ or $pd_a - \mathcal{L}$ in all source coefficients, not just first derivatives. Define normalized collision polynomials

$$F_o(S; z) = f_o(S; z) \prod_{i \in S} s_i(z)^{-1}. \tag{6}$$

The inverse exists in the finite nilpotent algebra. In particular, we do not divide by a possibly vanishing collision matrix element. The local matrix products and the sign are the finite CAR rule; no Poisson approximation has yet been made.

3 The fixed-word limit and its domination

Let $b(\pi)$ be the number of Hamiltonian–selected edges and let $\nu(\pi)$ count crossings plus nested pairs whose inner edge is selected. Only compatible forward diagrams survive. Define

$$K_\pi(z) = 2\sum_{i<j}\sum_{o=0,1} F_o(\{i,j\};z) - 2k(k-1) - 2k_G, \quad K_\pi(0) = -4\nu(\pi), \tag{7}$$

where k_G is the number of unrestricted edges. Put $q = e^{-4\lambda}$, $\gamma = e^{-\lambda} = q^{1/4}$ and $\kappa = 2\lambda(1-q)/(1+q)$. For r marks, the diagram contribution is

$$\gamma^{b(\pi)}[z_1\cdots z_r]\exp_{\mathrm{nil}}\left\{\lambda K_\pi(z) - \lambda\sum_{a:W} z_a - \kappa\sum_{a:j} z_a\right\}. \tag{8}$$

The constant exponential is $q^{\nu(\pi)}$; nilpotence applies to the remaining terms. Sum (8) over the allowed pairings. For normalized u, x, divide additionally by $\lambda^{n_W/2}v^{n_j/2}$, with $v = \lim v_N$.

Here is a fixed-degree convergence argument, including signed terms. Let $L_N = \binom{m}{p}^2$ and expand the exact coefficient of the empty and occupied site polynomials $f_0^m f_1^m$. After (5), assign $n_{o,S}$ collision sites of membership $|S| \geq 2$, with $d_i = \sum_{o,S\ni i} n_{o,S}$. The normalized combinatorial factor is

$$\frac{(m)_{U_0}(m)_{U_1}}{(m)_p^{2k}}\prod_i (p)_{d_i}\prod_{o,S}\frac{F_o(S;z)^{n_{o,S}}}{n_{o,S}!}, \tag{9}$$

where U_o counts the singleton and collision sites in occupancy o. The source-free counting is unchanged from the previous Slater proof. The coefficient norm of each F_o is bounded by a fixed B. Its falling-factorial estimate therefore bounds the absolute series by a constant times the product of exponential series with activities

$$B\alpha_{S,N}, \qquad \alpha_{S,N} = m(p/m)^{|S|}. \tag{10}$$

For fixed slots and marks their total is bounded. The $|S| \geq 3$ activity tends to zero, and the total-defect tail above R tends to zero uniformly as $R \to \infty$. This establishes domination before discarding higher intersections or taking source coefficients.

At fixed defects, the falling-factorial background is $e^{-2\lambda k(k-1)}$ and each double membership has activity $p^2/m \to 2\lambda$. The Hamiltonian normalization uses $D_N = \binom{m}{p}\binom{m+p}{p}$, with $D_N/L_N \to e^{2\lambda}$. There are $n_H = 2k_G + b$ Hamiltonian slots, giving the additional $e^{-\lambda n_H}$. These factors produce (7)–(8). For a backward selected edge a correct singleton is forbidden, forcing $d_i = p$ and hence a divergent number of defects. Its contribution vanishes by the same tail bound. The source maps preserve singleton parity and cannot restore that forbidden entry. All fixed moments, including those of $P_N^\dagger P_N$, converge and stay bounded. Their finite-N positivity passes to the limit.

4 The central-sector identification is a direct calculation

When all edges are G, the empty and occupied source polynomials satisfy $T(S;z) = (f_0(S;z) + f_1(S;z))/2$, where T is the old local normalized trace. Its singleton is $t_i = s_i/2$. Thus its normalized two-label collision ratio is

$$R_{ij}^{\mathrm{trace}}(z) = \frac{T(\{i,j\};z)}{t_i(z)t_j(z)} = 2\sum_{o=0,1} F_o(\{i,j\};z). \tag{11}$$

The tracial background is $-\binom{2k}{2} - k = -2k^2$. For all G, the background in (7) is also $-2k^2$. Therefore the entire source exponent, with every coefficient retained, is identical to the previous tracial one. This proves the central limit from its own finite coefficient expansion; it is not an inference from concentration of the full-Fock charge average.

For a covariant word, $\mathbb{E}$ of that word is scalar on $\bigwedge^m \mathbb{C}^N$. Its Slater expectation consequently equals its normalized sector trace exactly. The established endpoint cancellation of the whole two-slot D field and the one-slot $\mathcal{L}$ rule now apply to (11). They give the mixed-q tracial law for (h,u,x) with covariance the identity and crossing matrix

$$\mathsf{Q} = \begin{pmatrix} q & q & q^2 \\ q & q & q^2 \\ q^2 & q^2 & q^4 \end{pmatrix}. \tag{12}$$

This tracial law and its usual bounded representation were already used in the full-trace bilocal stage. What is new here is its direct central-sector derivation and its coupling to selected observables. In particular all central fixed Schatten moments of x_N are bounded.

Let μ_q be the unit-variance q-Hermite measure and H_n its monic polynomials, with squared norm $[n]_q!$. The two-energy measure of x_N converges to the established Mehler measure

$$\mathrm{d}\nu(E,E') = K_{q^2}(E,E')\,\mathrm{d}\mu_q(E)\,\mathrm{d}\mu_q(E'), \qquad K_r(E,E') = \sum_{n\ge 0} \frac{r^n H_n(E)H_n(E')}{[n]_q!}. \tag{13}$$

The finite measure is positive: $\nu_N = \mathbb{E}\,\mathrm{tr}_m \sum_{a,b} \Pi_a x_N \Pi_b x_N\, \delta_{(E_a,E_b)}$. All its polynomial moments follow from the central mixed-word limit. Higher energy moments provide tightness and uniform integrability; the compact limiting support determines the measure. Since $q^2 < 1$, (13) is absolutely continuous with respect to the product of the nonatomic q-Hermite measures. It has no mass on $E = E'$.

Portmanteau on the closed diagonal therefore gives

$$\|\mathcal{P}_{H_N} x_N\|_{2,m}^2 = \nu_N(E = E') \longrightarrow 0. \tag{14}$$

Pinching is contractive in every Schatten norm. Interpolating (14) with higher fixed central moments gives convergence to zero in every fixed finite $L^r(\mathrm{tr}_m)$, including all degenerate energy blocks. No long-time and large-N limits are interchanged.

5 Actual curvature and selected polynomial preparations

The previously established finite-N identity is

$$A_{0,N} = (1 - \mathcal{P}_{H_N})\big((2p - D)Z_2\big)\big|_m. \tag{15}$$

The omitted regression term in (2) is a function of H_m plus a scalar and is killed by $1 - \mathcal{P}_{H_N}$. Thus, exactly,

$$a_N := \frac{A_{0,N}}{C_m\sqrt{v_N}} = (1 - \mathcal{P}_{H_N})x_N. \tag{16}$$

This a_N denotes a normalized observable, not the preparation norm used in the preceding orbit-density draft. The physical $J_{2,N}$ defined there with the limiting regression coefficient differs from $C_m j_N$ by a coefficient tending to zero times Z_2, and a scalar proportional to $C - C_m$. Both differences vanish in every fixed central Schatten norm. Equation (16) itself needs no such asymptotic replacement.

Now allow P_N to be any fixed polynomial in $h_N, u_N, x_N, M_N, M_N^\dagger$. It has a fixed Gaussian degree, is stabilizer-covariant, and has bounded preparation norm by the source theorem. The preceding polynomial orbit-density result yields a constant C_P, independent of N, with

$$\|R_{P,N}\|_{L^2(\mathrm{tr}_m)} \le C_P, \qquad \mathbb{E}\,\|X_N P_N O\|^2 \le C_P\,\|X_N\|_{4,m}^2 \tag{17}$$

for every covariant X_N. Combining (14) and (16),

$$\mathbb{E}\,\|(a_N - x_N)P_N O\|^2 \longrightarrow 0. \tag{18}$$

The actual normalized neutral slope b_N also obeys $\mathbb{E}\,\|(b_N - u_N)P_N O\|^2 \to 0$ for this enlarged class of preparations: its already proved central tracial error is unchanged, and the new P_N now has the required bounded orbit density.

More generally, let W_N be a fixed covariant block built from h_N, a_N, b_N and fixed h_N-time phases. Tracial Hölder and telescoping first replace a_N, b_N by x_N, u_N in L^2. Equation (17) and polarization of P_N, Q_N then give

$$\varphi_N(P_N^\dagger W_N(h_N, a_N, b_N)Q_N) \longrightarrow \varphi(P^\dagger W(h, x, u)Q). \tag{19}$$

For the time phases, central Gaussian moments of h_N give a uniform Schatten Taylor remainder of order $S^n[(rn-1)!!]^{1/r}/n!$ at fixed even r. Each fixed number of covariant insertions has uniformly bounded higher norms, so Hölder gives L^2 truncation uniformly on compact real time windows. Apply the density bound before removing the truncation. This proves the time version of (19).

The selected factors in P, Q are unrestricted fixed polynomials. Several nonpolynomial exact-pinched blocks separated by selected factors still require an additional argument: their intervening preparations are not Gaussian polynomials. The present result does not replace that problem by convergence on a core.

6 Bounded covariant observables on the joint GNS space

The source theorem defines a positive functional on the full polynomial $*$-algebra. Form its GNS quotient and completion; let $\mathcal{C}$ be its dense polynomial core. The fields h, u, x are symmetric there, while $M^\dagger$ is the algebraic adjoint of M. We can identify the covariant action without guessing an independent selected representation.

Let $\mathcal{D}_0$ be the tracial mixed-q polynomial algebra in (12), with faithful trace τ, and let $A = A^*$ be one of its polynomials. For $\psi = P\Omega$, the finite-N density bound and both moment limits imply, for every fixed integer n,

$$\|A^n\psi\|^2 \le C_P \left[\tau(A^{4n})\right]^{1/2} \le C_P \|A\|_{\mathcal{D}}^{2n}. \tag{20}$$

Here $\mathcal{D}$ is the usual tracial C^* completion. The constant C_P does not grow with n; it is supplied by the fixed degree of the preparation, not the degree of the observable.

Log-convexity of the positive polynomial moments, or repeated Cauchy–Schwarz for $n = 2^k$, gives

$$\|A\psi\|^2 \le \|\psi\|^{2-2/n} \|A^n\psi\|^{2/n} \le \|\psi\|^{2-2/n} C_P^{1/n} \|A\|_{\mathcal{D}}^2. \tag{21}$$

Letting $n \to \infty$ proves $\|A\psi\| \le \|A\|_{\mathcal{D}} \|\psi\|$. Thus covariant null polynomials act as zero, all these operators extend boundedly, and the representation extends to $\mathcal{D}$. The vacuum restriction is τ, so it is faithful. In particular,

$$\|h\|, \|u\| \le \frac{2}{\sqrt{1-q}}, \qquad \|x\| \le \frac{2}{\sqrt{1-q^4}}. \tag{22}$$

There is also a useful normality statement. For every core vector,

$$|\langle\psi, A\psi\rangle| \le C_P \|A\|_{L^2(\tau)}. \tag{23}$$

Riesz representation gives an $L^2(\tau)$ density ρ_ψ for this functional. Positivity on polynomial squares extends by bounded approximation to positivity of the density; its trace is $\|\psi\|^2$. Hence these vector functionals are normal on the tracial von Neumann completion of $\mathcal{D}$. Polarization and density of $\mathcal{C}$ give a normal representation there.

This establishes a common representation of the actual covariant fields and the selected polynomial algebra. It does not classify the complete enlarged algebra, prove compactness of M on this larger space, or provide a closed open-chord transfer for j. Compactness in the older h, u, M cyclic subspace alone would not prove any of those statements.

7 Same-disorder mixed moments that distinguish the construction

The diagram rule gives the following unnormalized j moments:

$$\varphi(j^2) = v, \qquad \varphi(jh^2) = 0, \tag{24}$$

$$\varphi(j(M^\dagger)^2) = -2\kappa\gamma^2 q, \qquad \varphi(jM^\dagger h) = -2\kappa\gamma q, \tag{25}$$

$$\varphi(jhM^\dagger) = 0, \qquad \varphi(jWM^\dagger) = \varphi(jM^\dagger W) = 0,$$

$$\varphi(MjM^\dagger) = 0, \qquad \varphi(Mj^2M^\dagger) = v. \tag{26}$$

The order in (25) matters. To see its first identity directly, pair the two adjacent j slots with the two selected creations. Every surviving label assignment has disjoint occupied-to-empty hops. The unregressed loss deficit $2p - D$ on that pure $2p$-body hop is zero. The remaining regression term is $-\kappa\varphi(h^2(M^\dagger)^2)$, and the two forward pairings give $\varphi(h^2(M^\dagger)^2) = 2\gamma^2 q$. This finite-label reasoning agrees with the normalized source rule.

Since $\kappa > 0$ for every fixed $\lambda > 0$, the overlap is nonzero on the physical curve $\gamma = e^{-\lambda}$. A separately centered, disorder-independent probe would have zero such overlap. The tracial marginal law in (12) therefore cannot be used as a substitute for the common-disorder selected moments.

Equation (19) transfers these identities to actual curvature insertions. For example,

$$\lim_N \varphi_N(a_N(M_N^\dagger)^2) = -\frac{2\kappa\gamma^2 q}{\sqrt{v}}, \qquad \lim_N \varphi_N(M_N a_N^2 M_N^\dagger) = 1. \tag{27}$$

The second preparation already has unit norm. The first expression is an off-diagonal matrix element, not an expectation in the normalized double-excited state. Its adjoint gives the reversed matrix element. The physical curvature is recovered by multiplying by $C_m\sqrt{v_N}$ before taking the limit.

For an algebraic calibration, form the Gram matrix of the fourteen preparations

$$1,\ H,\ W,\ M^\dagger,\ H^2,\ HW,\ WH,\ W^2,\ M^\dagger H,\ HM^\dagger,\ M^\dagger W,\ WM^\dagger,\ (M^\dagger)^2,\ j. \tag{28}$$

Its exact LDL pivots are positive at the two saved rational choices $(\gamma, \lambda) = (1/2, 1/4), (2/3, 1/9)$. These are formal polynomial diagnostics, not points on $\gamma = e^{-\lambda}$ and not a proof of the full infinite-dimensional positivity or of a new orthogonal sector. General physical positivity comes from the finite-N construction and its moment limit.

8 Checks, sources and the next interface

The independent finite checker builds full-Fock integer CAR matrices for $(N, p) = (2, 1), (4, 1), (4, 2)$. It applies the global maps in (1) to entire blocks and compares every source coefficient with the product of the empty/occupied local polynomials. It includes forward, backward and unrestricted label restrictions, internal Gaussian pairs at this finite algebra level, and mixed $D/\mathcal{L}$ block layouts. The limiting driver separately enforces Wick exclusions. There are 592,704 exact comparisons, including 40 negative contributions. The maximum recorded integer magnitude is 16; the bounded-size matrix products are safely within integer range.

Further checks compare 326 all-G source exponents with the existing full-trace helper, their words with the established mixed-field formula, and 340 words without j with the older Slater $H/U/M$ transfer. Ten symbolic mixed expressions include (24)–(26); two fourteen-vector Gram checks use exact rational arithmetic. These finite comparisons test signs, source placement, counterterms and normalization. The arbitrary fixed-word convergence, bounded representation and exact pinching are proved above, not inferred from their finite coverage.

The source method builds on our already recorded full-trace multimark and bilocal derivations, and on the fixed-Slater defect expansion. The physical fixed-charge problem and the complex-SYK CAR/chord conventions have established precedents [1]. The dependence of the emergent algebra on retained state-adapted observables is also explicit in recent pure-state SYK work [2]; its Majorana/KM construction is not substituted for the present charge-conserving same-J composite calculation.

The next task is to turn the constructive selected-composite rule into an efficient transfer and identify its low-lying modules, then include the higher charged and neutral descendants of the full bilocal. The completed full-trace finite-time bilocal four-point construction remains available. The generalized source rule can treat fixed higher whole-block Wick marks, but their actual rank and charge-removal interfaces, uniform time tails in the selected state, and interlaced exact-pinching moments require separate work. The bulk spatial and mass generators and finite-G dynamics remain open. This stage is a microscopic observable construction, not a completed de Sitter duality.

References

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